

A. Spit Problem

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

In a Berland's zoo there is an enclosure with camels. It is known that camels like to spit. Bob watched these interesting animals for the whole day and registered in his notepad where each animal spitted. Now he wants to know if in the zoo there are two camels, which spitted at each other. Help him to solve this task.

The trajectory of a camel's spit is an arc, i.e. if the camel in position x spits d meters right, he can hit only the camel in position $x + d$, if such a camel exists.

Input

The first line contains integer n ($1 \leq n \leq 100$) — the amount of camels in the zoo. Each of the following n lines contains two integers x_i and d_i ($-10^4 \leq x_i \leq 10^4$, $1 \leq |d_i| \leq 2 \cdot 10^4$) — records in Bob's notepad. x_i is a position of the i -th camel, and d_i is a distance at which the i -th camel spitted. Positive values of d_i correspond to the spits right, negative values correspond to the spits left. No two camels may stand in the same position.

Output

If there are two camels, which spitted at each other, output YES. Otherwise, output NO.

Examples

input	Copy
2 0 1 1 -1	
output	Copy
YES	

input	Copy
3 0 1 1 1 2 -2	
output	Copy
NO	

input	Copy
5 2 -10 3 10 0 5 5 -5 10 1	
output	Copy
YES	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 29 (Div. 2, Codeforces format)

Finished

Practice



→ Virtual participation

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Language:

Choose file: [Choose File](#) No file chosen

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Submission	Time	Verdict
373215009	May/01/2026 12:52	Accepted
373213569	May/01/2026 12:30	Accepted
373213391	May/01/2026 12:28	Accepted



→ **Problem tags**

brute force *1000

No tag edit access

→ **Contest materials**

- Announcement 
- Tutorial (en) 

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